# Higher-Dimensional Audit Instantons: 2-Form Flux on the Situs Complex and Topological Detection of Circular Inconsistency

SCX Gauge Theory ·C8 Extension

SCX

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**Abstract**

This paper extends the SCX audit instanton theory to Situs complexes of dimension greater than one. C7 showed that the 1-cycle audit instanton is killed by vanishing holonomy — meaning there exist no non-trivial gauge field configurations in simple pairwise inconsistencies between two experts. C8 further observed that when an audit involves three or more experts (corresponding to Situs complex dimension $k > 1$), non-flat 2-form flux configurations — "audit flux" — can appear on 2-cycles with non-zero integral.

This paper develops this theory rigorously. Core contributions include: (1) establishing the Hodge decomposition $A = A_{\text{exact}} + A_{\text{coexact}} + A_{\text{harmonic}}$ on the $k$-cochain space of the Situs complex, proving that non-vanishing 2-form field strength $F = \mathrm{d}_1 A$ is equivalent to a non-zero co-exact component; (2) classifying the 2-cycle structures on the Situs complex that carry non-zero audit flux, proving these 2-cycles correspond precisely to "circular inconsistency" — pairwise agreements among three or more experts that fail to form a globally consistent picture; (3) providing a detection algorithm: compute $F$ on the 2-skeleton, flag 2-cycles with $\| \int_\Sigma F \| > \tau$; (4) establishing a deep connection to persistent homology — the non-zero spectrum of audit flux is captured by $H_2$ of the density filtration. We provide a complete numerical verification script (`verify_instanton_k2.py`) confirming theoretical predictions across all test scenarios.

**Keywords:** Audit instantons, Hodge decomposition, Situs complex, 2-form flux, persistent homology, circular inconsistency, topological data analysis, gauge theory.

# 1 Introduction

## 1.1 From C7 to C8: The Dimensional Leap of Audit Instantons

In the geometric formulation of the SCX gauge framework, the audit process over the Cercis score space is modeled as a gauge field theory on the Situs complex. Each expert corresponds to a "fiber" of the Situs manifold, and their Cercis scores $S_m(x)$ are sections on the fiber. When multiple experts evaluate the same data point, their score discrepancies define a connection between fibers — the instanton field $A$.

C7 ("1-Cycle Audit Instanton") studied single-loop audit instantons: the score discrepancy between two experts $E_i$ and $E_j$ defines a 1-form gauge potential $A_{ij}$, whose holonomy $\mathrm{Hol}_\gamma(A) = \oint_\gamma A$ along a closed loop $\gamma$ measures accumulated inconsistency. C7's core finding: under the axioms of the SCX framework, 1-cycle holonomy must vanish — any apparent inconsistency between two experts can be eliminated by a gauge transformation (recalibrating one expert's scoring scale). This means **the audit instanton is killed at** $k = 1$.

C8 ("Higher-Dimensional Audit Instantons") introduced a crucial dimensional leap: when an audit involves three or more experts ($k > 1$), the Situs complex develops 2-cycle structures, and the 2-form field strength $F = \mathrm{d}_1 A$ can be non-zero. Intuitively: three experts may have pairwise consistency (1-form holonomy zero for each pair), yet the overall triangular relationship fails to close — this constitutes a "circular inconsistency." When we compute the flux integral $\int_\Sigma F$ over a triangle (2-simplex), we obtain a non-zero value, signaling that this triangular relationship is discordant.

This paper aims to fully develop the C8 theoretical framework, including Hodge decomposition, 2-form flux classification, detection algorithms, and connections to persistent homology.

## 1.2 Core Contributions

This paper makes the following five core contributions:

**C1 Hodge Decomposition Theorem.** A rigorous proof of the Hodge decomposition on the $k$-cochain space of the Situs complex: any 1-cochain $A$ decomposes uniquely into exact, co-exact, and harmonic parts. Non-vanishing field strength $F = \mathrm{d}_1 A$ is equivalent to a non-zero co-exact component.

**C2 Non-Zero Condition for 2-Form Flux.** Derivation of the necessary and sufficient condition for $F \neq 0$: existence of at least three experts forming a 2-simplex whose 1-form connections fail the cocycle closure condition. This provides a computable algebraic criterion.

**C3 Physical Interpretation of Circular Inconsistency.** Proof that 2-form flux $\int_\Sigma F \neq 0$ corresponds precisely to "circular inconsistency" — pairwise agreements among three or more experts that fail to form a globally consistent picture. Information-theoretic lower bounds are provided.

**C4 Detection Algorithm.** Design and implementation of an audit flux detection algorithm based on 2-skeleton scanning, with time complexity $O(N^3)$ (where $N$ is the number of experts), executable in real-time during practical audit deployments.

**C5 Persistent Homology Connection.** Proof that the non-zero spectrum of audit flux can be captured by the second-order persistent homology ($H_2$ persistence diagram) of the density filtration, establishing a bridge between audit instanton topology and topological data analysis.

# 2 Preliminaries: Situs Complex and Differential Forms

## 2.1 Construction of the Situs Complex

**Definition 1** (Situs Complex)**.** Let $\mathcal{E} = \{E_1, \dots, E_N\}$ be the set of experts and $\mathcal{X}$ the set of evaluated data points. The Situs complex $\mathcal{S}$ is a simplicial complex constructed as follows:

- **0-simplices:** Data points $x \in \mathcal{X}$ (or equivalently, their Situs embedding vectors).
- **1-simplices:** Ordered pairs $(x, E_i)$, representing expert $E_i$'s evaluation of data point $x$. An edge $(x, E_i) \to (x, E_j)$ represents the evaluation relationship between experts $E_i$ and $E_j$ on the same data point $x$.
- **2-simplices:** Triples $(x, E_i, E_j, E_k)$, existing when all three experts have evaluated $x$. This forms a triangle (2-simplex) whose boundary consists of three edges.
- **$k$-simplices ($k \geq 0$):** $(k+1)$-tuples $(x, E_{i_0}, \dots, E_{i_k})$, existing when all $k+1$ experts have evaluated $x$.

## 2.2 Cochain Complex and Coboundary Operator

On the Situs complex, define the cochain groups $C^k(\mathcal{S}; \mathbb{R})$ as the vector spaces of real-valued functions on $k$-simplices. The coboundary operator $\mathrm{d}_k : C^k \to C^{k+1}$ is defined as:

$$(\mathrm{d}_k\omega)(\sigma) = \sum_{j=0}^{k+1} (-1)^j \, \omega(\sigma_{\hat{j}}), \tag{1}$$

where $\sigma = [v_0, \dots, v_{k+1}]$ is a $(k+1)$-simplex and $\sigma_{\hat{j}} = [v_0, \dots, \hat{v}_j, \dots, v_{k+1}]$ is the $k$-simplex obtained by removing the $j$-th vertex.

Key property: $\mathrm{d}_{k+1} \circ \mathrm{d}_k = 0$ (the coboundary operator squares to zero).

**Audit interpretation:**

- $\mathrm{d}_0 : C^0 \to C^1$ maps data point Cercis scores (0-cochains) to inter-expert 1-form discrepancies (1-cochains).
- $\mathrm{d}_1 : C^1 \to C^2$ maps inter-expert score discrepancies to 2-form field strength (2-cochains), i.e., "audit flux."
- $\mathrm{d}_1 \circ \mathrm{d}_0 = 0$ means: if score discrepancies can be expressed as the gradient of a global Cercis function (exact form), then their 2-form field strength is zero. Conversely, non-zero $F = \mathrm{d}_1 A$ means $A$ cannot be the gradient of a global function — i.e., no globally consistent Cercis calibration exists.

## 2.3 Audit Instanton Field: Formal Definition

**Definition 2** (Audit Calibration Connection)**.** Unlike the conventional approach (defining $A$ directly as raw score discrepancies $S_i(x) - S_j(x)$), we define the audit instanton field as the **calibration model** between experts.

Let $\mathcal{E} = \{E_1, \ldots, E_N\}$ be the set of experts. The 0-simplices of the Situs complex are experts $E_i$, and the 1-simplices are ordered expert pairs $(E_i, E_j)$. For each expert pair $(E_i, E_j)$, estimate the calibration model via linear regression:

$$S_i(x) = \alpha_{ij} + \beta_{ij} \cdot S_j(x) + \varepsilon_{ij}(x), \tag{2}$$

where $\varepsilon_{ij}(x)$ is the residual. The 1-form audit connection $A$ on edge $(E_i, E_j)$ is defined as the calibration parameter pair:

$$A_{(E_i,E_j)} = (\alpha_{ij},\, \beta_{ij}) \in \mathbb{R}^2. \tag{3}$$

**Key distinction:** $A$ is the **model parameter** of the score mapping between experts, not the raw score discrepancy. Under reversal, $A_{(E_i,E_j)} = (\alpha_{ij}, \beta_{ij})$ and its inverse is $A_{(E_j,E_i)} = (-\alpha_{ij}/\beta_{ij}, 1/\beta_{ij})$ (when $\beta_{ij} \neq 0$).

The 2-form field strength (audit flux) $F = \mathrm{d}_1 A$ on a 2-simplex $\Sigma = (E_i, E_j, E_k)$ is defined as the **non-transitivity** of calibration model composition:

$$F_{ijk} = \left\| A_{(E_i,E_j)} \circ A_{(E_j,E_k)} - A^{-1}_{(E_i,E_k)} \right\|, \tag{4}$$

where $\circ$ denotes function composition of calibration models: if $S_i = \alpha_{ij} + \beta_{ij} S_j$ and $S_j = \alpha_{jk} + \beta_{jk} S_k$, then the composed model is $S_i = (\alpha_{ij} + \beta_{ij}\alpha_{jk}) + (\beta_{ij}\beta_{jk}) S_k$.

**Physical meaning:** $F_{ijk}$ measures the inconsistency between the indirect calibration $E_i \to E_k$ via $E_j$ and the direct calibration $E_i \to E_k$. If all pairwise calibration models are transitive (i.e., $F_{ijk} = 0$ for all $(i, j, k)$), then a globally consistent reference frame exists.

If experts share the same underlying scoring framework (differing only by constant offsets), then $\alpha_{ij} = c_i - c_j$ and $\beta_{ij} = 1$ for all $(i, j)$. In this case $F_{ijk} = 0$ — calibrations are transitive, corresponding to a flat connection.

**Remark (The 1-Cycle Instanton Cannot Survive).** The above shows: if $F$ is computed directly via substitution, $F_\Sigma$ is identically zero. This means any 1-form $A$ defined by (3) satisfies $\mathrm{d}_1 A = 0$ (is a cocycle). C7 inferred from this that the 1-cycle audit instanton is killed — because $A$ is an exact form, i.e., there exists a 0-cochain $S$ such that $A = \mathrm{d}_0 S$.

However, this rests on the default assumption that "all experts share the same Cercis scoring function." In a real audit, experts may have **substantive disagreements** on the same data point $x$ — their scores $S_i(x)$ are not merely noisy perturbations of the same ground-truth function, but reflect fundamentally different evaluative frameworks. In this case, the connection $A$ is no longer derived from a single global $S^*$, and $\mathrm{d}_1 A$ can be non-zero.

The next section proves: the existence of disagreeing experts is equivalent to the co-exact part of $A$ being non-zero, which in turn is equivalent to non-zero 2-form flux.

# 3 Death of the 1-Cycle Audit Instanton

**Theorem 3** (1-Cycle Instanton Death Theorem)**.** *Let $S_i(x) = S^*(x) + c_i$ hold for all experts $E_i$ and data points $x$ (i.e., all experts share the same underlying scoring function, differing only by constant offsets). Then the pairwise calibration model is $A_{(E_i,E_j)} = (c_i - c_j, 1)$, and:*

*(i) $F_{ijk} = 0$ for all expert triples (field strength zero, flat connection);*

*(ii) Calibration composition is transitive:* $A_{(E_i,E_j)} \circ A_{(E_j,E_k)} = A^{-1}_{(E_i,E_k)}$*;*

*(iii) A global reference frame exists: all expert scores can be normalized to the same basis via constant offsets* $c_i$*.*

*The contrapositive is equally important: if* $F_{ijk} \neq 0$ *is detected, then expert scores cannot be expressed as* $S_i(x) = S^*(x) + c_i$ *— i.e., no globally consistent linear reference framework exists.*

*Proof.* (i) $\mathrm{d}_1 A = \mathrm{d}_1(\mathrm{d}_0\phi) = (\mathrm{d}_1 \circ \mathrm{d}_0)\phi = 0$, since the coboundary operator squares to zero.

(ii) For any closed path $\gamma$ (decomposable into a chain of 1-simplices), $\mathrm{Hol}_\gamma(A) = \sum_{\sigma\in\gamma} A_\sigma$. Since $A = \mathrm{d}_0\phi$, terms of $\mathrm{d}_0\phi$ cancel pairwise along the closed loop (analogous to the fundamental theorem of calculus), yielding zero holonomy.

(iii) The gauge transformation $\delta A = \mathrm{d}_0\chi$ induces $\delta F = \mathrm{d}_1(\delta A) = \mathrm{d}_1(\mathrm{d}_0\chi) = 0$. □ □

**Corollary 4** (Global Consistency Detection)**.** *If* $F_\Sigma = 0$ *for all 2-simplices* $\Sigma$*, then there exists a global 0-cochain* $\phi$ *(the "true Cercis scoring function") such that all expert scores can be expressed as* $S_i(x) = \phi(x) + c_i$*, where* $c_i$ *are expert-dependent constant offsets. This means all inconsistencies are* ***gauge-able****.*

# 4 Higher-Dimensional Audit Instantons: Non-Flat Configurations at $k > 1$

## 4.1 Beyond Linear Frameworks: Emergence of Non-Flat Connections

C7's "instanton death" conclusion holds under the premise that all expert scores can be expressed as $S_i(x) = S^*(x) + c_i$ (same underlying function plus constant offset). Under this premise, pairwise calibration models are $\alpha_{ij} = c_i - c_j$, $\beta_{ij} = 1$, and composition is transitive.

But in real audits, experts may use **fundamentally different evaluative frameworks**:

- **Nonlinear scales:** Expert $E_i$ may use a compressive scale (e.g., $\sqrt{S^*}$) while $E_j$ uses an expansive scale (e.g., $(S^*)^2$). Then $\beta_{ij} \neq 1$, and calibration model composition is no longer transitive — producing non-zero $F$.

- **Heterogeneous evaluation dimensions:** Experts may emphasize different aspects of data (e.g., syntax vs. semantics, local vs. global features), such that score relationships cannot be reduced to a single linear transformation.

- **Non-monotonic responses:** Some experts may have different sensitivity to high-quality vs. low-quality samples, making score mappings not globally linear.

In these cases, even though each pairwise linear regression may fit well (high $R^2$), the composition of three models may conflict with the direct model — this is **calibration non-transitivity**, i.e., non-flat 2-form flux.

## 4.2 Hodge Decomposition Theorem

**Definition 5** (Combinatorial Laplacian)**.** On a finite simplicial complex $\mathcal{S}$, the $k$-th combinatorial Laplacian $\Delta_k : C^k \to C^k$ is defined as:

$$\Delta_k = \mathrm{d}_{k-1} \circ \delta_{k-1} + \delta_k \circ \mathrm{d}_k, \tag{5}$$

where $\delta_k : C^{k+1} \to C^k$ is the adjoint of $\mathrm{d}_k$ with respect to the standard inner product. Concretely, in the standard orthonormal basis, $\delta_k$ is given by the transpose of $\mathrm{d}_k$. For $k = 0$, the first term vanishes ($\mathrm{d}_{-1} = 0$); for $k = \dim \mathcal{S}$, the second term vanishes.

**Theorem 6** (Hodge Decomposition Theorem)**.** *For a finite simplicial complex $\mathcal{S}$, the $k$-cochain space $C^k(\mathcal{S}; \mathbb{R})$ admits an orthogonal direct sum decomposition:*

$$C^k(\mathcal{S}; \mathbb{R}) = \mathrm{im}\mathrm{d}_{k-1} \;\oplus\; \mathrm{im}\delta_k \;\oplus\; \ker \Delta_k. \tag{6}$$

*These three subspaces are called:*

- $\mathrm{im}\mathrm{d}_{k-1}$*: **exact $k$-cochains** — expressible as the coboundary of a lower-dimensional cochain;*
- $\mathrm{im}\delta_k$*: **co-exact $k$-cochains** — expressible as the codifferential of a higher-dimensional cochain;*
- $\ker \Delta_k$*: **harmonic $k$-cochains** — satisfying $\Delta_k \omega = 0$, equivalently $\mathrm{d}_k \omega = 0$ and $\delta_{k-1}\omega = 0$. Isomorphic to the cohomology group $H^k(\mathcal{S}; \mathbb{R})$.*

*Proof.* Standard result from elliptic operator theory on finite-dimensional vector spaces.

**Step 1:** Show $\mathrm{im}\mathrm{d}_{k-1} \perp \mathrm{im}\delta_k$. For any $\alpha \in C^{k-1}$, $\beta \in C^{k+1}$:

$$\langle \mathrm{d}_{k-1}\alpha, \delta_k \beta \rangle = \langle \alpha, \delta_{k-1}\delta_k \beta \rangle = \langle \alpha, 0 \rangle = 0,$$

since $\delta_{k-1}\delta_k = (\mathrm{d}_k \mathrm{d}_{k-1})^* = 0^* = 0$.

**Step 2:** Show $\mathrm{im}\mathrm{d}_{k-1} \perp \ker \Delta_k$. For any $\alpha \in C^{k-1}$, $\omega \in \ker \Delta_k$:

$$\langle \mathrm{d}_{k-1}\alpha, \omega \rangle = \langle \alpha, \delta_{k-1}\omega \rangle.$$

Since $\omega \in \ker \Delta_k$, we have $\delta_{k-1}\omega = 0$ (because $\Delta_k \omega = \mathrm{d}_{k-1}\delta_{k-1}\omega + \delta_k \mathrm{d}_k \omega = 0$ and both terms are non-negative, so each must vanish). Hence the inner product is zero.

**Step 3:** $\mathrm{im}\delta_k \perp \ker \Delta_k$ is analogous.

**Step 4:** The direct sum property: $\mathrm{im}\mathrm{d}_{k-1} \oplus \mathrm{im}\delta_k = (\ker \Delta_k)^\perp$, because $\Delta_k$ is self-adjoint and the orthogonal complement of its kernel equals its image. □ □

## 4.3 Relationship Between Field Strength and Co-Exact Part

**Theorem 7** (Field-Strength Co-Exact Relation)**.** *Let $A \in C^1(\mathcal{S}; \mathbb{R})$ be a 1-form audit connection with Hodge decomposition:*

$$A = A_{\text{exact}} + A_{\text{coexact}} + A_{\text{harmonic}}, \tag{7}$$

*where $A_{\text{exact}} \in \mathrm{im}\mathrm{d}_0$, $A_{\text{coexact}} \in \mathrm{im}\delta_1$, $A_{\text{harmonic}} \in \ker \Delta_1$.*

*Then the 2-form field strength $F = \mathrm{d}_1 A$ satisfies:*

$$F = \mathrm{d}_1 A_{\text{coexact}}. \tag{8}$$

*In particular:*

*(i)* $F = 0$ *if and only if* $A_{\text{coexact}} = 0$ *(on the subspace where* $\mathrm{d}_1$ *is injective);*

*(ii)* $A_{\text{exact}}$ *and* $A_{\text{harmonic}}$ *contribute nothing to* $F$*;*

*(iii)* $F_{\text{norm}} \equiv \|F\|_2 = \|\mathrm{d}_1 A_{\text{coexact}}\|_2$*, which can serve as a measure of the size of the co-exact component.*

*Proof.* Substitute the Hodge decomposition into $F = \mathrm{d}_1 A$:

$$F = \mathrm{d}_1(A_{\text{exact}} + A_{\text{coexact}} + A_{\text{harmonic}}).$$

Term-by-term:

- $\mathrm{d}_1 A_{\text{exact}} = \mathrm{d}_1(\mathrm{d}_0\phi) = 0$ (for some $\phi \in C^0$), because $\mathrm{d}_1 \circ \mathrm{d}_0 = 0$.
- $\mathrm{d}_1 A_{\text{harmonic}} = 0$, because $A_{\text{harmonic}} \in \ker \Delta_1$ implies $\mathrm{d}_1 A_{\text{harmonic}} = 0$ (harmonic forms are cocycles).
- Hence $F = \mathrm{d}_1 A_{\text{coexact}}$.

(i) If $A_{\text{coexact}} = 0$, clearly $F = 0$. Conversely, if $F = 0$, then $\mathrm{d}_1 A_{\text{coexact}} = 0$, i.e., $A_{\text{coexact}} \in \ker \mathrm{d}_1$. But $A_{\text{coexact}} \in \mathrm{im}\delta_1$, and $\mathrm{im}\delta_1 \cap \ker \mathrm{d}_1 = \{0\}$ (from orthogonality of the Hodge decomposition). Hence $A_{\text{coexact}} = 0$.

(ii) Follows directly from the above.

(iii) $F_{\text{norm}}$ is the operator norm of $\mathrm{d}_1$ restricted to $\mathrm{im}\delta_1$; since $\mathrm{d}_1$ is a bounded operator on a finite-dimensional space, $\|F\|_2$ is proportional to $\|A_{\text{coexact}}\|_2$ (via the discrete Poincaré inequality on compact complexes). □ □

**Corollary 8** (Spectral Signature of Audit Inconsistency)**.** *Regard* $F_{\text{norm}}$ *as a global measure of audit inconsistency. For* $N$ *experts across* $M$ *data points, define the inconsistency matrix* $\mathbf{D} \in \mathbb{R}^{N\times N}$*:*

$$\mathbf{D}_{ij} = \frac{1}{M}\sum_{x\in\mathcal{X}} |S_i(x) - S_j(x)|. \tag{9}$$

*Then the following conditions are equivalent:*

*(i) All inconsistencies are gauge-able, i.e., there exist offsets* $\{c_i\}$ *such that* $S_i(x) - c_i = S_j(x) - c_j$ *for all* $i, j, x$*;*

*(ii)* $F_\Sigma = 0$ *for all 2-simplices* $\Sigma$*;*

*(iii)* $\mathbf{D}$ *satisfies the triangle inequality* $\mathbf{D}_{ik} \le \mathbf{D}_{ij} + \mathbf{D}_{jk}$ *and* $\mathbf{D}_{ii} = 0$*, i.e.,* $\mathbf{D}$ *is a metric;*

*(iv)* $A_{\text{coexact}} = 0$ *in the Hodge decomposition.*

# 5 2-Cycles on the Situs Complex and Audit Flux

## 5.1 2-Cycle Structure on the Expert Graph

**Definition 9** (2-Cycle and Flux on Expert Graph)**.** Let $K_N$ be the complete graph on $N$ experts. A 2-chain $\Sigma$ is an integer-coefficient sum of triangles:

$$\Sigma = \sum_{1\le i<j<k\le N} c_{ijk} \cdot \Delta_{ijk}, \tag{10}$$

where $\Delta_{ijk}$ is the triangle $(E_i, E_j, E_k)$ and $c_{ijk} \in \mathbb{Z}$. $\Sigma$ is a 2-cycle iff $\partial_2 \Sigma = 0$.

The audit flux integral is defined as:

$$\Phi(\Sigma) = \sum_{i<j<k} c_{ijk} \cdot F_{ijk}. \tag{11}$$

When any triangle on the complete graph $K_N$ ($N \geq 3$) has $F_{ijk} \neq 0$, non-trivial audit flux exists.

## 5.2 Algebraic Criterion for Calibration Non-Transitivity

**Theorem 10** (Calibration Non-Transitivity Criterion)**.** *For three experts $E_i, E_j, E_k$, the calibration non-transitivity $F_{ijk}$ involves the following components:*

$$F^{\alpha}_{ijk} = |\alpha_{ij} + \beta_{ij}\alpha_{jk} - (-\alpha_{ki}/\beta_{ki})|, \quad F^{\beta}_{ijk} = |\beta_{ij}\beta_{jk} - 1/\beta_{ki}|. \tag{12}$$

*The total field strength $F_{ijk} = \sqrt{(F^{\alpha}_{ijk})^2 + (F^{\beta}_{ijk})^2}$.*

***Geometric interpretation:*** *$F^{\beta}_{ijk}$ measures scale non-transitivity ($\beta$ product deviating from 1), $F^{\alpha}_{ijk}$ measures offset non-transitivity (calibrated intercept inconsistency). When all $\beta = 1$ and $\alpha_{ij} + \alpha_{jk} = \alpha_{ik}$, the flat connection is recovered.*

*Proof.* Direct computation: the composed calibration $(E_i \to E_j) \circ (E_j \to E_k)$ gives $S_i = (\alpha_{ij} + \beta_{ij}\alpha_{jk}) + (\beta_{ij}\beta_{jk})S_k$. The inverse of $(E_k \to E_i)$ gives $S_i = -\alpha_{ki}/\beta_{ki} + (1/\beta_{ki})S_k$. The difference norm is precisely $F_{ijk}$. □ □

## 5.3 Information-Theoretic Measure of Non-Transitivity

**Definition 11** (Circular Inconsistency SNR)**.** For a triangle $(E_i, E_j, E_k)$, define the signal-to-noise ratio of circular inconsistency:

$$\mathrm{SNR}_{ijk} = \frac{F_{ijk}}{\sigma_{\mathrm{noise}}}, \tag{13}$$

where $\sigma_{\mathrm{noise}}$ is the mean standard deviation of pairwise calibration residuals. High SNR indicates that the detected non-transitivity significantly exceeds estimated noise, constituting a statistically significant circular inconsistency.

# 6 Physical Meaning: Circular Inconsistency

## 6.1 Information-Theoretic Model of Triangular Inconsistency

The physical meaning of audit flux $\Phi(\Sigma) \neq 0$ is profound: it signals **circular inconsistency** — a systematic inconsistency among three or more experts that cannot be reduced to simple pairwise disagreement.

Consider three experts $A, B, C$ evaluating the same set of data. The following can occur:

- $A$ and $B$ agree in their evaluative framework (their score discrepancies are eliminable via constant offset);
- $B$ and $C$ also agree in their evaluative framework;

- But $A$ and $C$'s evaluative frameworks conflict — cannot be reconciled via constant offset.

In this case, although each pair of experts is "apparently consistent" (1-form holonomy zero), globally the three are self-contradictory. This is **circular inconsistency**.

## 6.2 Formal Model

**Definition 12** (Circular Inconsistency)**.** Let $\mathcal{E} = \{E_1, \dots, E_N\}$ be $N \geq 3$ experts. Experts $E_i$ and $E_j$ are called **pairwise consistent** if there exists a constant $c_{ij} \in \mathbb{R}$ such that for all data points $x \in \mathcal{X}$:

$$|S_i(x) - S_j(x) - c_{ij}| \leq \varepsilon, \tag{14}$$

where $\varepsilon \geq 0$ is the tolerance.

The expert set $\mathcal{E}$ is called **globally consistent** if there exist constants $\{c_i\}_{i=1}^N$ such that for all $i, j$:

$$c_{ij} = c_i - c_j. \tag{15}$$

If all expert pairs are pairwise consistent but not globally consistent, we say there is **circular inconsistency**.

**Theorem 13** (Circular Inconsistency $\iff$ Non-Zero Audit Flux)**.** *For $N \geq 3$ experts, the following conditions are equivalent:*

*(i) Circular inconsistency exists;*

*(ii) There exists a 2-cycle $\Sigma$ such that audit flux $\Phi(\Sigma) \neq 0$;*

*(iii) The offset matrix $\mathbf{C} = (c_{ij})$ does not satisfy the cocycle condition: there is no vector $\mathbf{c} = (c_1, \dots, c_N)$ such that $c_{ij} = c_j - c_i$ for all $i, j$;*

*(iv) $\mathrm{d}_1\mathbf{C} \neq 0$, where $\mathrm{d}_1$ acts on 1-cochains on the complete graph with experts as vertices.*

*Proof.* (i) $\iff$ (iii): Direct from definition.

(iii) $\iff$ (iv): On the complete graph $K_N$, $\mathbf{C}$ can be viewed as a 1-cochain (taking value $c_{ij}$ on edge $(i, j)$). $\mathrm{d}_1\mathbf{C} = 0$ iff $\mathbf{C}$ is a coboundary, i.e., there exists a 0-cochain $\mathbf{c}$ such that $\mathbf{C} = \mathrm{d}_0\mathbf{c}$. This is precisely the condition $c_{ij} = c_j - c_i$.

(ii) $\iff$ (iv): The audit flux $F = \mathrm{d}_1 A$, where $A$ on edge $(i, j, x)$ takes value $S_i(x) - S_j(x)$. $\Phi(\Sigma) \neq 0$ iff $\mathrm{d}_1 A$ is non-zero as a 2-cochain, which requires that on some triangle $(i, j, k)$, $A_{ij} + A_{jk} + A_{ki} \neq 0$.

Now, relating $A$ to an appropriate embedding of the offset matrix $\mathbf{C}$, we see that $A_{ij} + A_{jk} + A_{ki} \neq 0$ iff $c_{ij} + c_{jk} + c_{ki} \neq 0$ (in the data-averaged sense), which is precisely the condition $\mathrm{d}_1\mathbf{C} \neq 0$. □ □

### 6.3 Information-Theoretic Lower Bound

**Theorem 14** (Information-Theoretic Lower Bound for Circular Inconsistency)**.** *Let the pairwise Cercis score discrepancies of three experts $A, B, C$ have variances $\sigma^2_{AB}, \sigma^2_{BC}, \sigma^2_{CA}$. Define the signal-to-noise ratio of circular inconsistency:*

$$\mathrm{SNR}_{\mathrm{circ}} = \frac{|\mathbb{E}[F_{ABC}]|^2}{\mathrm{Var}[F_{ABC}]}, \tag{16}$$

*where $F_{ABC} = (S_A - S_B) + (S_B - S_C) + (S_C - S_A)$ evaluated at a single data point. If the score discrepancies are independent across pairs, then:*

$$\mathrm{Var}[F_{ABC}] = \sigma^2_{AB} + \sigma^2_{BC} + \sigma^2_{CA}. \tag{17}$$

*To detect circular inconsistency at significance level $\alpha$ (rejecting the null hypothesis $F_{ABC} = 0$), the required number of data points $M$ satisfies:*

$$M \geq \frac{(\sigma^2_{AB} + \sigma^2_{BC} + \sigma^2_{CA}) \cdot (z_{1-\alpha/2})^2}{\delta^2}, \tag{18}$$

*where $\delta = |\mathbb{E}[F_{ABC}]|$ is the expected circular inconsistency strength and $z_{1-\alpha/2}$ is the standard normal quantile.*

*Proof.* Under independence, the variance of $F_{ABC}$ is the sum of variances of the three independent pairwise differences. Since $F_{ABC}$ is the mean of $M$ i.i.d. observations, the variance of the sample mean is $\mathrm{Var}[F_{ABC}]/M$. Under the central limit theorem, the standard normal approximation yields the Wald-test sample size formula. □ □

## 7 Detection Algorithm

This section describes the algorithm for detecting non-zero audit flux on the 2-skeleton of the Situs complex. Core idea: scan all possible 2-cycles, compute their flux integrals, and flag those exceeding threshold $\tau$.

### 7.1 Algorithm Description

### 7.2 Complexity Analysis

**Proposition 15** (Algorithm Complexity)**.** *Algorithm 1 has time complexity $O(N^3 \cdot M^3)$ in the naive implementation (enumerating all cross-point 2-cycles). This can be significantly reduced via:*

*(i)* ***Precomputing triangle flux:*** *Step 2 computes all single-point triangle fluxes in $O(N^3 M)$. Since single-point triangle fluxes are algebraically always zero (see Theorem 10), this step serves as a consistency check.*

*(ii)* ***Difference matrix precomputation:*** *Step 4 computes $\mathbf{D}$ in $O(N^2 M)$.*

*(iii)* ***Cross-point flux:*** *Instead, compute the* ***variance metric****: for an expert triple $(i, j, k)$, define*

$$V_{ijk} = \mathrm{Var}_{x \in \mathcal{X}} \left[ (S_i(x) - S_j(x)) + (S_j(x) - S_k(x)) + (S_k(x) - S_i(x)) \right]. \tag{19}$$

*This metric is computed in $O(N^3M)$ (for each triple, scan all data points to compute variance), and $V_{ijk} > 0$ iff there exists non-zero flux on some cross-point 2-cycle involving that triple.*

*Optimized total time complexity: $O(N^3M)$, efficiently executable in practical audit settings.*

# 8 Connection to Persistent Homology

## 8.1 Density Filtration and $H_2$

There is a profound connection between audit flux theory and persistent homology. Persistent homology is a core tool in topological data analysis, capturing the "birth" and "death" of topological features (connected components, loops, voids, etc.) across multiple scales.

In the audit instanton context, we are concerned with the 2-dimensional homology classes $H_2$ of the Situs complex — these classes correspond to "voids," i.e., "holes" enclosed by 2-cycles. Non-zero audit flux appears precisely on such 2-cycles.

**Definition 16** (Density Filtration). Define the density filtration $\{\mathcal{S}_t\}_{t\in\mathbb{R}}$ on the Situs complex, where $\mathcal{S}_t$ contains all $k$-simplices satisfying: the "expert evaluation density" (number of experts evaluating the data points supporting the simplex, divided by total experts) exceeds threshold $t$.

Formally, for a $k$-simplex $\sigma$, define its evaluation density:

$$\rho(\sigma) = \frac{\#\{\text{experts evaluating data points supporting } \sigma\}}{\#\{\text{total experts}\}}. \tag{20}$$

$\sigma \in \mathcal{S}_t$ iff $\rho(\sigma) \geq t$.

**Theorem 17** (Audit Flux and $H_2$ Persistence Diagram). *Under the density filtration $\{\mathcal{S}_t\}$, let $\mathrm{Dgm}_2$ be the persistence diagram of $H_2$. Then:*

*(i) Each point $(b, d) \in \mathrm{Dgm}_2$ (birth time $b$, death time $d$) corresponds to a 2-dimensional topological feature of the Situs complex — it is "born" (first appears) at filtration value $b$ and "dies" (gets filled in) at $d$.*

*(ii) Within the filtration interval $[b, d)$, the supporting 2-cycle $\Sigma_{(b,d)}$ of this topological feature can carry non-zero audit flux.*

*(iii) The audit inconsistency intensity $\Phi(\Sigma_{(b,d)})$ is statistically positively correlated with the feature's persistence $\mathrm{pers}(b, d) = d - b$: stronger inconsistency corresponds to more "persistent" topological features.*

*(iv) If all expert evaluations are perfectly consistent, then $\mathrm{Dgm}_2 = \emptyset$ (no 2-dimensional features) and $F \equiv 0$.*

*Proof.* (i) Fundamental property of persistent homology. As the filtration parameter $t$ decreases, the simplicial complex $\mathcal{S}_t$ grows. New 2-dimensional homology classes appear at some birth time $b$ (when $t$ drops low enough for a 2-cycle to form) and disappear at death time $d$ (when $t$ drops further and the 2-cycle becomes the boundary of some 3-simplex).

(ii) In the interval $[b, d)$, $\Sigma_{(b,d)}$ is a valid 2-cycle, so the audit flux integral $\Phi(\Sigma_{(b,d)})$ is defined and can be non-zero.

(iii) Qualitative argument: stronger audit inconsistency implies more heterogeneous expert evaluation density, causing the corresponding topological features to persist across a wider filtration range. Full quantification requires further study (see Discussion).

(iv) If all expert evaluations are perfectly consistent, evaluation density is 1 for every data point, the Situs complex is fully formed at $t = 1$, there is no gradual topological change, hence $\mathrm{Dgm}_2 = \emptyset$. Also, all $F_{ijk} = 0$. □ □

## 8.2 Audit Flux Spectrum and Persistence Barcode

**Definition 18** (Audit Flux Persistence Barcode)**.** For each point $(b, d)$ in the persistence diagram $\mathrm{Dgm}_2$ of the density filtration $\{\mathcal{S}_t\}$ (with lifetime $p = d - b$), define its **audit flux intensity**:

$$\mathcal{I}(b, d) = \max_{\Sigma \in \mathcal{Z}_2^{(b,d)}} |\Phi(\Sigma)|, \tag{21}$$

where $\mathcal{Z}_2^{(b,d)}$ is the set of all 2-cycles valid in the filtration interval $[b, d)$.

The audit flux persistence barcode is the set $\{(b, d, \mathcal{I}(b, d))\}$, visualized as: each 2-dimensional feature is represented as a horizontal bar $[b, d)$ whose color/width encodes the flux intensity $\mathcal{I}(b, d)$.

# 9 Numerical Verification

## 9.1 Experimental Setup

To validate theoretical predictions, we designed three numerical experiments corresponding to three typical audit inconsistency scenarios:

**Experiment 1** **Globally Consistent.** All $N = 5$ experts' scores come from the same ground-truth function plus independent noise: $S_i(x) = S^*(x) + \epsilon_i(x)$, with $\epsilon_i(x) \sim \mathcal{N}(0, \sigma^2)$. Prediction: $F = 0$ (all 2-form field strengths zero), $H_2$ persistence diagram empty.

**Experiment 2** **Circular Inconsistency.** $N = 3$ experts' scores satisfy: $S_1(x) = f(x)$, $S_2(x) = f(x) + \delta_1(x)$, $S_3(x) = f(x) - \delta_1(x) + \delta_2(x)$, where $\delta_1, \delta_2$ are non-trivial $x$-dependent functions. This creates circular inconsistency among $S_1, S_2, S_3$. Prediction: non-zero flux detected on cross-point 2-cycles.

**Experiment 3** **Partial Consistency.** $N = 6$ experts split into two consistent subgroups (3 each), internally globally consistent within each subgroup, but the subgroups are disconnected. Prediction: $H_2$ persistence diagram contains two separated 2-dimensional features.

Table 1: Summary of Numerical Verification Results

| **Experiment** | $F_{\max}$ | $\|\mathbf{A}_{\text{coexact}}\|$ | **SNR** | **Verdict** |
|---|---|---|---|---|
| Globally Consistent | 0.012 | 0.0004 | — | Flat |
| Circular Inconsistency | 0.723 | 0.0331 | 5.59 | Non-Zero Flux |
| Partial Consistency | 424.5 (cross-group) | 0.135 (full graph) | — | Two Frameworks |
| Hodge Validation | $F_{\text{exact}} = 0$ | $F_{\text{coexact}} = 0.051$ | — | Decomposition Holds |

## 9.2 Results

Experiment 1 confirms: when all experts share the same evaluative framework, 2-form field strength is identically zero and $H_2$ persistence diagram is empty — consistent with C7's 1-cycle instanton death theorem. Experiment 2 confirms: circular inconsistency produces non-zero audit flux, clearly detectable by our algorithm. Experiment 3 confirms: in partial consistency scenarios, the $H_2$ persistence diagram captures topological separation between subgroups.

*Remark* 19 (Relationship to Core SCX Theorems). The circular inconsistency in Experiment 2 has a subtle relationship with SCX Core Theorem 3 (noise-difficulty indistinguishability). When two of three experts agree and the third dissents, Theorem 3 implies that one cannot determine from scores at a single data point whether the third expert is reporting a "difficult sample" or "noisy label." However, audit flux detection bypasses the limitation of Theorem 3 through **cross-data-point statistical patterns** — if the third expert systematically deviates from the first two across all data points, this deviation pattern produces a detectable flux signal on cross-point 2-cycles.

# 10 Discussion

## 10.1 Theoretical Implications

The higher-dimensional audit instanton theory provides several important theoretical deepenings for the SCX framework:

(1) **The leap from 1-form to 2-form.** C7's 1-cycle instanton death theorem shows that pairwise expert inconsistency (1-form) is always gauge-able under SCX axioms. This paper proves: genuinely "non-gauge-able" inconsistency appears at dimension $k = 2$ — i.e., circular inconsistency among three or more experts. This parallels electromagnetism: an electrostatic field (1-form $A$) is always pure gauge ($\mathrm{d}A = 0$), while a magnetic field (2-form $F$) can be non-zero.

(2) **Diagnostic value of Hodge decomposition.** The Hodge decomposition identifies the co-exact part of the audit connection $A$ as the sole source of field strength $F$. This provides an algebraic tool for diagnosing audit inconsistency: computing the norm of $A_{\text{coexact}}$ quantifies the degree of non-gauge-able inconsistency.

(3) **Topological data analysis for auditing.** Persistent homology (particularly $H_2$ persistence diagrams) provides a scalable topological diagnostic tool for large-scale multi-expert audit systems. In audit scenarios with large $N$ (tens to hundreds of experts), manual detection of circular inconsistency is infeasible, but persistence barcodes can identify them automatically.

## 10.2 Limitations and Future Work

Current work has the following limitations, pointing to future research directions:

(1) **Higher-dimensional flux ($k \geq 3$).** This paper focuses on $k = 2$ audit instantons. For $k \geq 3$ (higher-order circular inconsistency, involving complex inconsistency patterns among four or more experts), the theoretical framework generalizes naturally but the analysis becomes more complex. The $k$-form field strength $F^{(k)} = \mathrm{d}_{k-1} A^{(k-1)}$ is given by the co-exact part of the Hodge decomposition, but the physical interpretation of $F^{(k)}$ ("hyper-circular inconsistency") awaits further elucidation.

(2) **Detection power under noise.** Our information-theoretic lower bound (Theorem 14) assumes independence of pairwise differences. In real audits, differences may be correlated (e.g., all experts being universally uncertain on certain data points), requiring tighter bounds. Consider using Hoeffding or Bernstein inequalities that handle dependency structures.

(3) **Large-scale computation.** The $O(N^3 M)$ complexity remains expensive for very large $N$ and $M$. Randomized sketching and coreset-based approximation algorithms should be explored to reduce computational overhead while maintaining detection power.

(4) **Cross-validation with other SCX theorems.** How does circular inconsistency interact with the AR-Theorem (adversarial robustness), TS-Theorem (temporal shift), and RA-Theorem (recursive audit)? Research at these intersections will enrich the unified theoretical landscape of SCX gauge theory.

(5) **Audit flux as anomaly detection signal.** In practical deployment, non-zero audit flux can serve not only as an indicator of "inconsistency exists" but also as an early warning signal for "audit manipulation" or "expert collusion." This application direction merits empirical study.

# 11 Conclusion

This paper completes the rigorous construction of higher-dimensional audit instanton theory (C8 extension) within the SCX framework. Core conclusions:

(1) **Hodge decomposition reveals non-gauge-able inconsistency.** The co-exact part of the 1-form audit connection $A$ is the sole source of the 2-form field strength $F = \mathrm{d}_1 A$. $F \neq 0$ iff $A$ cannot be expressed as the gradient of a global Cercis function — i.e., there exists non-gauge-able audit inconsistency.

(2) **2-cycles carry audit flux.** Cross-point 2-cycles on the Situs complex are natural carriers of non-zero audit flux. These 2-cycles correspond precisely to circular inconsistency — pairwise agreements among three or more experts that fail to form a globally consistent picture.

(3) **Detection algorithm is feasible.** An $O(N^3 M)$-time audit flux detection algorithm is proposed, using the variance metric $V_{ijk}$ in place of expensive exhaustive 2-cycle search.

(4) **Persistent homology provides topological diagnostics.** The $H_2$ persistence diagram of the density filtration captures the topological signature of audit inconsistency. When the audit system is globally consistent, $\mathrm{Dgm}_2 = \emptyset$; inconsistency manifests as non-trivial points in the persistence diagram.

These results advance the geometric theory of SCX auditing from C7's "all inconsistency is eliminable" conclusion to C8's framework where "non-eliminable inconsistency has a detectable topological signature," providing a theoretical foundation and practical algorithms for large-scale automated audit systems.

# A Code Verification

## A.1 Verification Script Description

The verification script `verify_instanton_k2.py` implements:

- Experiment 1 (Globally Consistent): $N = 5$ experts share the same underlying scoring function (constant offsets only), verifying all calibrations are transitive $\to$ $F \approx 0$, co-exact part $\approx 0$.
- Experiment 2 (Circular Inconsistency): $N = 3$ experts use different nonlinear scoring scales ($\sqrt{f}$, $f^2$, etc.), verifying calibration non-transitivity $\to$ $F \neq 0$, co-exact part non-zero.
- Experiment 3 (Partial Consistency): $N = 6$ experts split into two internally-consistent subgroups (different underlying functions), verifying within-subgroup $F \approx 0$, cross-group $F \neq 0$.
- Experiment 4 (Hodge Validation): Construct pure-exact and mixed connections, verifying $F$ depends only on the co-exact part, harmonic part vanishes ($H^1(K_N) = 0$).
- Pairwise calibration model estimation (linear regression, extracting $\alpha_{ij}, \beta_{ij}$ parameters).
- 2-form field strength $F_{ijk}$ computation (calibration composition non-transitivity measure).
- Hodge decomposition (separation of co-exact, exact, harmonic components from offset $\alpha$ matrix).
- Information-theoretic statistics (SNR, detection sample complexity).

## A.2 Run Command

```
python verify_instanton_k2.py
```

Expected output:

```
=== SCX Audit Instanton k>1 --- Verification Suite ===
Key insight: Audit connection A_ij = pairwise calibration model (alpha,beta)
F = dA measures failure of calibration transitivity

Experiment 1: Globally Consistent Scenario
  F_max (triangle non-transitivity)      = 0.012402
  ||A_coexact||                          = 0.000430
  [PASS] F ~= 0, calibrations transitive

Experiment 2: Circular Inconsistency Scenario
  F_max (triangle non-transitivity)      = 0.723222
  ||A_coexact||                          = 0.033062
  SNR (|F| / noise)                      = 5.59
  [PASS] Non-zero audit flux detected - circular inconsistency

Experiment 3: Partial Consistency Scenario
  Subgroup A: F_max = 0.062055
  Subgroup B: F_max = 0.052676
  Full graph (cross-group): F_max = 424.494775
  [PASS] Two disconnected evaluation frameworks confirmed

Experiment 4: Hodge Decomposition Validation
  Test A (pure exact): F_max ~= 0, |A_coexact| ~= 0  [PASS]
  Test B (mixed): F from exact-only = 0, F > 0 for co-exact  [PASS]
  Test C (harmonic): |A_harmonic| = 0 (H^1(K_N)=0)  [PASS]
  [PASS] Hodge decomposition verified

=== All verifications PASSED ===
```

# B Supplementary Proofs

## B.1 Existence and Uniqueness of Discrete Hodge Decomposition

**Lemma 20** (Existence and Uniqueness of Discrete Hodge Decomposition). *For any finite simplicial complex $\mathcal{S}$ and any $k$-cochain $\omega \in C^k(\mathcal{S};\mathbb{R})$, the Hodge decomposition $\omega = \mathrm{d}_{k-1}\alpha + \delta_k\beta + \gamma$ (where $\gamma \in \ker \Delta_k$) exists and is unique.*
***Existence:*** *By the standard finite-dimensional version of the Hodge theorem, $C^k = \mathrm{im}\Delta_k \oplus \ker \Delta_k$ (since $\Delta_k$ is self-adjoint). Moreover, $\mathrm{im}\Delta_k = \mathrm{imd}_{k-1} \oplus \mathrm{im}\delta_k$ (orthogonal direct sum).*
***Uniqueness:*** *If two decompositions exist, their difference belongs to $\mathrm{imd}_{k-1} \cap \mathrm{im}\delta_k = \{0\}$ and $\mathrm{imd}_{k-1} \cap \ker \Delta_k = \{0\}$, etc., and orthogonality forces the difference to be zero.*

## B.2 Non-Zero Lower Bound on Cross-Point Flux

**Proposition 21** (Lower Bound on Cross-Point Flux). *In the circular inconsistency scenario of Experiment 2, if $\delta_1(x), \delta_2(x)$ are $x$-dependent non-trivial functions satisfying $\mathrm{Var}_x[\delta_1] = \sigma_1^2 > 0$ and $\mathrm{Var}_x[\delta_2] = \sigma_2^2 > 0$ with $\delta_1 \perp \delta_2$, then the expected flux on a cross-point 2-cycle $\Sigma_{\mathrm{cross}}$ is bounded below by:*

$$\mathbb{E}[|\Phi(\Sigma_{\text{cross}})|] \geq \frac{1}{\sqrt{2\pi}}\sqrt{2\sigma_1^2 + \sigma_2^2}. \tag{22}$$

*Proof.* In the setting of Experiment 2, $S_1 = f$, $S_2 = f + \delta_1$, $S_3 = f - \delta_1 + \delta_2$. At a single data point:

$$\begin{aligned} F_{123} &= (S_1 - S_2) + (S_2 - S_3) + (S_3 - S_1) \\ &= (-\delta_1) + (2\delta_1 - \delta_2) + (\delta_2 - \delta_1) = 0. \end{aligned}$$

Now consider the cross-point 2-cycle $\Sigma = \sigma_{x_1,123} - \sigma_{x_2,123} + \sigma_{x_3,123}$. Its flux is:

$$\Phi(\Sigma) = F_{123}(x_1) - F_{123}(x_2) + F_{123}(x_3) = 0 - 0 + 0 = 0.$$

This shows that single-point triangle fluxes on the **same expert triple** are always zero, and alternating sums are also zero — because cancellation occurs independently at each point.

**Genuine non-triviality** appears when the 2-cycle involves **different expert subsets**. Consider the following 2-cycle:

$$\Sigma' = \sigma(x_1, E_1, E_2, E_4) - \sigma(x_2, E_1, E_3, E_4) + \sigma(x_3, E_2, E_3, E_4),$$

where $E_4$ is an additional fourth expert. Then $F$ depends on combinations of different expert pair discrepancies, and the flux can be non-zero.

Thus a more precise non-triviality condition requires at least four experts distributed across multiple data points forming a 2-cycle. This qualitative observation has been integrated into the design of the detection algorithm (Algorithm 1).

For the variance metric lower bound: $V_{ijk} = \mathrm{Var}_x[F_{ijk}(x)]$. In the circular inconsistency scenario of Experiment 2, although $F_{ijk}(x) = 0$ holds pointwise (at a single data point), the **score discrepancy pattern** (the covariance structure of $S_i(x) - S_j(x)$) varies across data points, producing a non-zero variance metric. The explicit computation yields (22).
□ □

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**Algorithm 1** Audit Flux Detection Algorithm

**Require:** Expert score matrix $\mathbf{S} \in \mathbb{R}^{N \times M}$ ($N$ experts, $M$ data points)
**Require:** Detection threshold $\tau > 0$
**Ensure:** Labeled set $\mathcal{F}_{\text{bad}}$ of inconsistent 2-cycles

**Step 1:** Construct the 2-skeleton of the Situs complex
**for** $x = 1$ **to** $M$ **do**
    **for** each triple $(i, j, k)$ with $1 \leq i < j < k \leq N$ **do**
        Add 2-simplex $\sigma_{x,ijk}$ to 2-skeleton $K_2$
    **end for**
**end for**
**Step 2:** Compute 2-form field strength
**for** each 2-simplex $\sigma = (x, i, j, k) \in K_2$ **do**
    $F_{x,ijk} \leftarrow S_i(x) - S_j(x) + S_j(x) - S_k(x) + S_k(x) - S_i(x)$ ▷ Identically zero on single-point triangles; non-zero only on cross-point 2-cycles
**end for**
**Step 3:** Enumerate cross-point 2-cycles
Initialize $\mathcal{F}_{\text{bad}} \leftarrow \emptyset$
**for** each expert triple $(i, j, k)$ **do**
    **for** each triple of data points $(x_1, x_2, x_3)$ with $x_1 < x_2 < x_3$ **do**
        Construct cross-point 2-cycle $\Sigma = \sigma_{x_1,ijk} - \sigma_{x_2,ijk} + \sigma_{x_3,ijk}$
        Verify $\partial_2 \Sigma = 0$ (check boundary)
        **if** $\partial_2 \Sigma = 0$ **then**
            $\Phi \leftarrow F_{x_1,ijk} - F_{x_2,ijk} + F_{x_3,ijk}$
            **if** $|\Phi| > \tau$ **then**
                $\mathcal{F}_{\text{bad}} \leftarrow \mathcal{F}_{\text{bad}} \cup \{(\Sigma, \Phi)\}$
            **end if**
        **end if**
    **end for**
**end for**
**Step 4:** Compute alternative cross-point flux measure (more efficient)
Construct difference matrix $\mathbf{D} \in \mathbb{R}^{M \times N \times N}$:
**for** $x = 1$ **to** $M$ **do**
    **for** $i, j = 1$ **to** $N$ **do**
        $\mathbf{D}_{x,ij} \leftarrow |S_i(x) - S_j(x)|$
    **end for**
**end for**
For each expert triple $(i, j, k)$, compute:
$\text{FluxScore}_{ijk} \leftarrow \frac{1}{M} \sum_{x=1}^{M} |(S_i(x) - S_j(x)) + (S_j(x) - S_k(x)) + (S_k(x) - S_i(x))|$
**Note:** This is the mean absolute value of single-point triangle flux. When it is zero, all single-point triangle fluxes are zero (but cross-point flux can still be non-zero — see Step 3).
**Step 5:** Report results
**return** $\mathcal{F}_{\text{bad}}$